

Smart Real-Time Age, Gender, Emotion and Behavior - Based Advertising System for Sri Lankan Malls

Using Custom-Built Models



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PP2 Panel Comment	Actions Taken
Need to conduct more validations	Conducted additional model evaluations using real-world testing scenarios and benchmark metrics
Better if you can gather real-world statistics	Deployed in a simulated retail environment and on AWS for real-time access.
Need to justify day/night fear classification	Fear during daytime may occur in normal crowded situations, while nighttime fear is treated with higher priority to reduce false alerts and improve security sensitivity.
Need stronger understanding of Ad recommendation component	Refined each module independently and prepared detailed explanations for model architecture and decision logic



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Problem Statement

Traditional digital advertising systems show generic advertisements without understanding customer demographics, emotions, or engagement.

Existing System Limitations

- No real-time customer analysis
- No personalized advertisement recommendations
- Limited engagement and emotion analysis
- Weak support for returning customer recognition
- Reduced customer interaction effectiveness

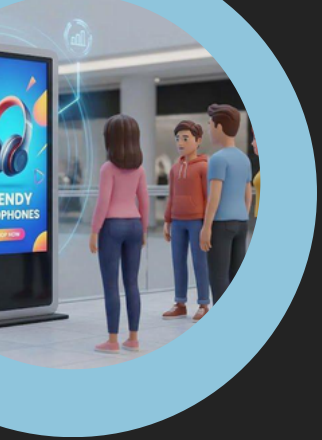
Proposed Solution

- **Real-time Age, Gender, Emotion detection and engagement analysis**
- **Personalized advertisement recommendation while detecting suspicious activities for mall security.**
- **Dual operating modes with kiosk-friendly real-time analytics support**



INTERACTIVE GAME-BASED AGE AND GENDER DETECTION FOR TARGETED ADVERTISING

B.Sc. (Hons) in Information Technology specializing in Data Science



INTERACTIVE GAME-BASED AGE AND GENDER DETECTION FOR TARGETED ADVERTISING

A real-time age and gender detection system using CNN and MTCNN to support personalized advertisement targeting in live kiosk environments.

Metric	Your Model	DeepFace	OpenCV
Age Accuracy	0.93	0.37	0.45
Age F1 Score	0.93	0.27	0.40
Gender Accuracy	0.91	0.79	0.75
Gender F1 Score	0.91	0.78	0.75

Validation & Key Findings

- Custom CNN outperformed DeepFace and OpenCV
- Improved age and gender prediction accuracy
- Reliable demographic detection in real-time environments
- Lightweight architecture enabled stable live performance



EMOTION DETECTION WITH DOMAIN-SPECIFIC CLASSIFICATION FOR ADAPTIVE ADS AND SECURITY ALERTS

B.Sc. (Hons) in Information Technology specializing in Data Science



EMOTION DETECTION WITH DOMAIN-SPECIFIC CLASSIFICATION FOR ADAPTIVE ADS AND SECURITY ALERTS

A context-aware emotion detection system that identifies six emotions in real time while separating security alerts and advertisement logic for accurate and reliable analysis.

Metric	MobileNetV2	Custom CNN
Accuracy	0.53	0.71
Macro F1 Score	0.50	0.70
Weighted F1 Score	0.50	0.71
Suspicious Emotion F1 Score	0.89	0.98
Fear Emotion F1 Score	0.18	0.70

Validation & Key Findings

- Custom CNN outperformed MobileNetV2
- Improved Emotion detection accuracy in real-time environments
- Better emotion class balance
- Reduced false alerts using context-awareness



REAL-TIME BEHAVIOR-DRIVEN ENGAGEMENT ANALYTICS AND ANOMALY INTEGRATION FOR ADAPTIVE ADS

B.Sc. (Hons) in Information Technology specializing in Data Science



REAL-TIME CONTEXT-AWARE BEHAVIOR-DRIVEN ENGAGEMENT ANALYTICS AND ANOMALY INTEGRATION FOR ADAPTIVE ADS

A real-time engagement detection system that analyzes attention, head orientation, and dwell time to measure customer interaction and identify behavioral anomalies.

Metric	ResNet	Custom CNN
Accuracy	0.82	0.86
Macro F1 Score	0.81	0.86
Weighted F1 Score	0.82	0.86
Looking at Kiosk F1 Score	0.77	0.84
Looking Away F1 Score	0.85	0.88

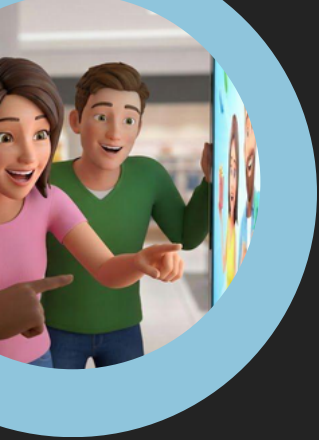
Validation & Key Findings

- Improved engagement detection accuracy in real-time environments
- Better identification of attentive and non-attentive users
- Successfully supported anomaly-aware engagement monitoring



DEMOGRAPHY-BASED ADVERTISEMENT RECOMMENDATION ENGINE INCORPORATING SERENDIPITY AND DEMOGRAPHIC FILTERING

B.Sc. (Hons) in Information Technology specializing in Data Science



DEMOGRAPHY-BASED ADVERTISEMENT RECOMMENDATION ENGINE INCORPORATING SERENDIPITY AND DEMOGRAPHIC FILTERING

A real-time personalized advertisement recommendation system that generates image and video ads using age, gender, emotion, and engagement analytics.

Validation Area	Purpose
Model File Validation	Verified required model files before loading
Data Structure Validation	Checked advertisement database schema
Text Validation	Normalized and validated input text
Model Loading Validation	Verified successful model loading
Scoring Validation	Evaluated weighted recommendation scoring

Validation & Key Findings

- Personalized ads improved user interaction
- Dynamic recommendations increased engagement potential
- System adapted effectively to changing user emotions
- Smooth integration achieved across all AI modules

Results and Performance



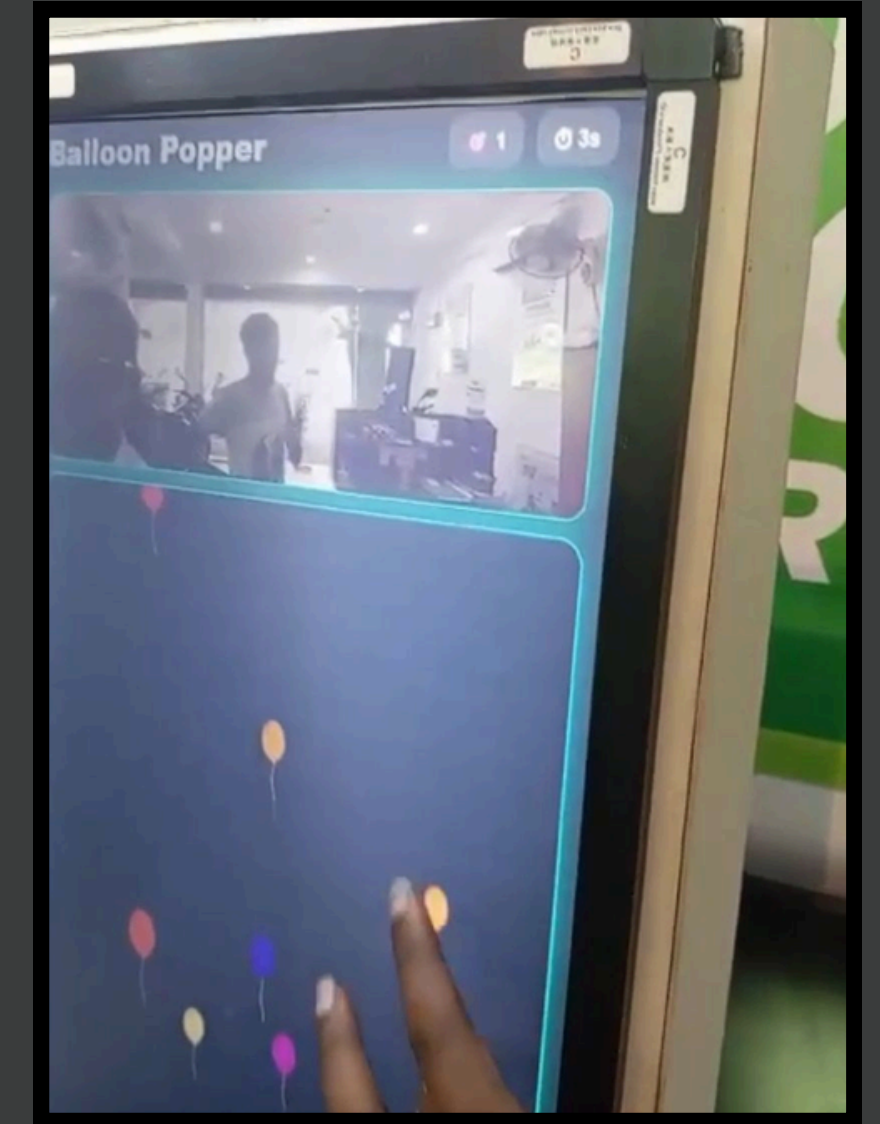
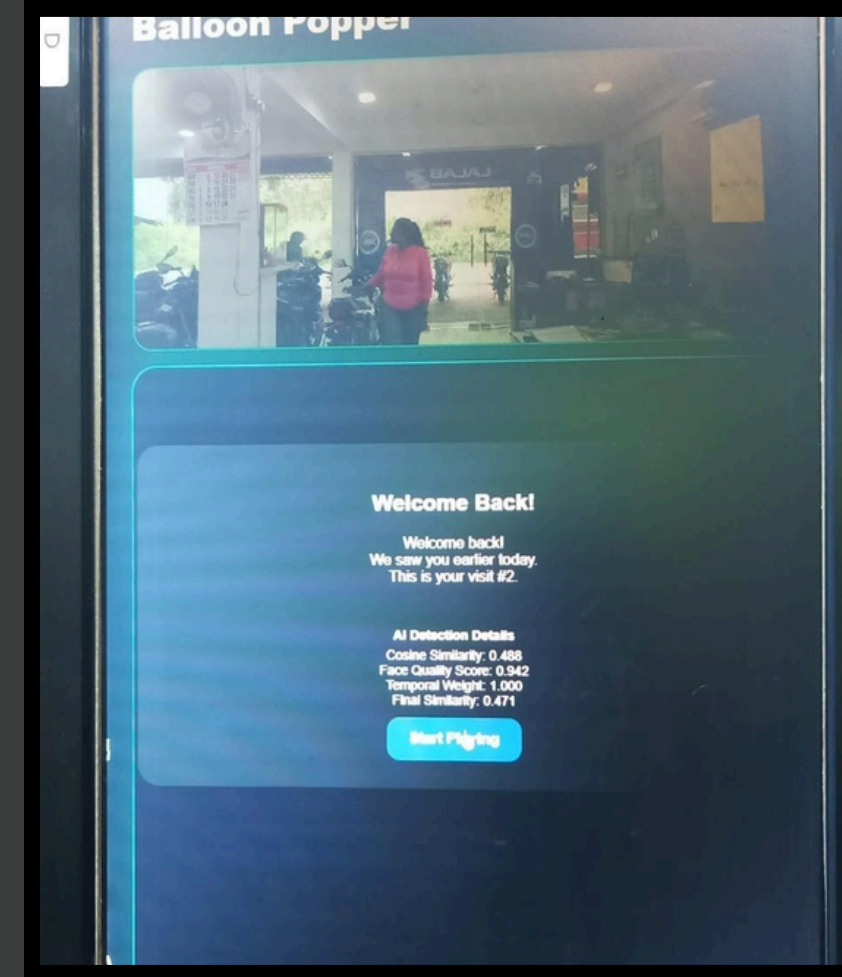
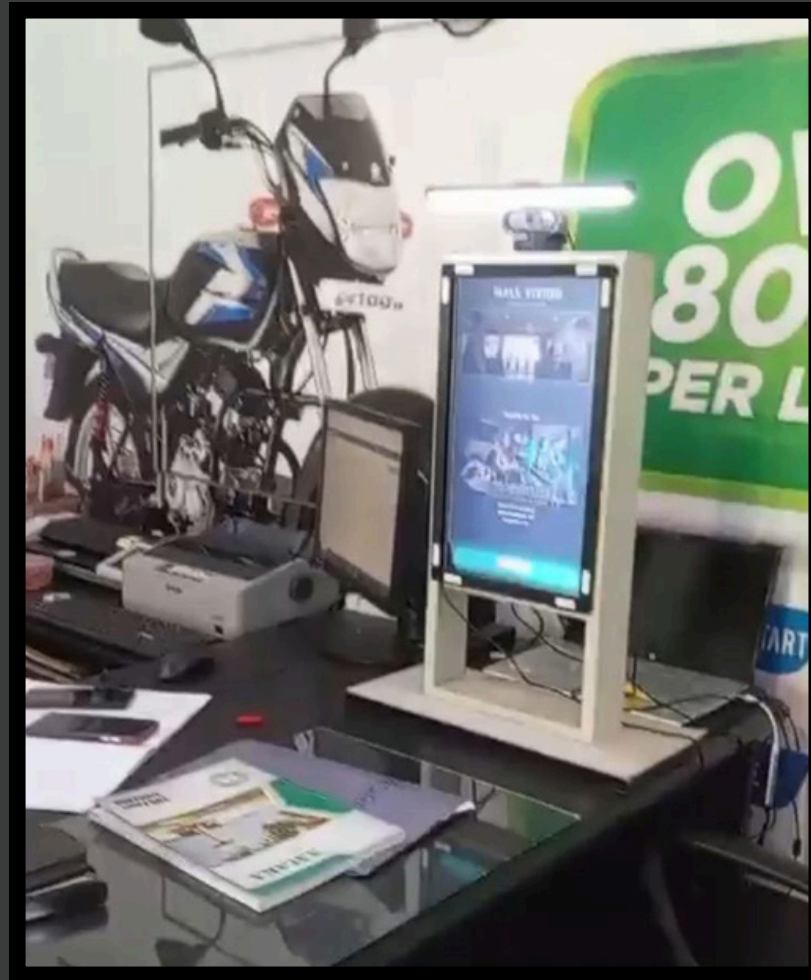
Model Performance

Module	Result
Age /Gender Classification	87%
Emotion Recognition	71%
Audience Detection	70%
Engagement Analysis	87%
Advertisement Recommendation Engine	72%

Validation and Real-World Testing

- Conducted testing in simulated retail kiosk environments
- Collected real-world interaction statistics
- Evaluated system responsiveness under live conditions
- Performed benchmark comparisons with baseline models

User Testing Sessions



User Feedback

Positive Observations

- Interactive experience increased user engagement
- Personalized advertisements improved user interest
- System operated smoothly in real-time scenarios

Limitations

- Performance may vary under poor lighting conditions
- Real-world deployment requires stable hardware resources

Future Improvements

- Advanced recommendation algorithms
- Multi-camera customer tracking
- Improved low-light performance



Thank you !